C-XGBoost model with targeted regularization for treatment effect estimation

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Abstract

Estimation of treatment effects constitutes a critical aspect of research, especially in safety-critical fields such as medicine, economics and health-care in which the uncertainty and estimation is vital for decision-making. By quantifying the impact of interventions, researchers are able to assess the causal relationships between treatments and outcomes, enabling more informed decision-making. In this research, we propose an extended version of C-XGBoost model for estimating the potential outcomes as well as the propensity score. A remarkable advantage of the proposed model is that it is trained with a new loss function, which is equipped with targeted regularization procedure that is based on non-parametric estimation theory and aims at reducing bias and improving estimation accuracy. The experimental analysis on three challenging collections of causal inference benchmarks provide strong empirical evidence about the efficiency of the proposed model while the statistical analysis secures the effectiveness and robustness of the proposed approach.

Keywords: Causal inference, treatment effect, C-XGBoost, targeted regularization

1 Introduction

The growing availability of observational data and the continued advances in artificial intelligence and machine learning offer new opportunities for advancing research in the estimation of treatment effects [1–3]. The main objective of treatment effect estimation (also referred as causal effect estimation) is the accurate approximation of the impact of an intervention, which is also known as "*treatment*", to a specific outcome. This process poses a significant challenge across various fields, such as medicine, finance and engineering [4] since only the outcome of the chosen intervention is observable, while other potential outcomes remain unknown (counterfactuals).

Let us consider, the d-dimensional space of covariates $\mathcal{X}$, the domain of outcomes $\mathcal{Y}$ and the joint distribution Π on $\mathcal{X} \times \{0,1\} \times \mathcal{Y}$. Given a dataset $(\mathbf{x}_i, t_i, y_i)$, where $\mathbf{x}_i \in \mathcal{X}$, $t_i \in \{0,1\}$, $y_i \in \mathcal{Y}$ and $i = 1, 2, \ldots, n$, the primary aim is the estimation of heterogeneous treatment effect, which is defined by

$$\psi = E[Y \mid X, T = 1] - E[Y \mid X, T = 0], \tag{1}$$

where the random variables X, T, Y corresponding to the covariates, treatment and outcome, respectively and $(X, T, Y) \sim \Pi$. Notice that $T = 0$ and $T = 1$ corresponds to the instances belonging to the control and treatment group. We point out that for every sample of the dataset $\mathbf{x}_i$, either $t_i = 0$ (namely $y_i^{(0)}$ has been observed) or $t_i = 1$ ($y_i^{(1)}$ has been observed) and $y_i = t_i y_i^{(1)} + (1 - t_i) y_i^{(0)}$ [5]. Notice that the observed outcome is often called *factual* while the corresponding unobserved is called *counterfactual*.

During the last decade, a variety of machine learning models and strategies were proposed for estimating treatment effect. The significant progress in neural networks and deep learning revolutionized the development of causal inference models [6–8] which demonstrated advanced prediction ability of the potential outcomes as well as the capability of simultaneously predicting the propensity score $g(x) = P(T = 1 \mid x)$ (i.e., the probability of an instance to be assigned to the treatment group). An additional attribute of neural network-based models is that their architecture includes a common backbone and separate heads for the estimation of both potential outcomes. This enables networks to learn a covariate representation that does not directly depend on the treatment, making them less vulnerable to misspecification and overfitting.

Recently, Kiriakidou and Diou [9] conducted a comprehensive performance evaluation of the most effective tree-based and neural network-based models on several collections of causal inference datasets and concluded that there is no class of models, which can be universally outperform the rest; hence, the model selection is the subject to investigation, depending on the dataset characteristics. In addition, neural network-based models are significantly impacted by uninformative features [10] and tend to favor overly smooth solutions and are biased towards low-frequency functions [11]. In contrast, tree-based model are not affected by uninformative features [10] and excel at capturing irregular patterns in the data (non-smooth target functions) due to their ability to learn piece-wise constant functions [11]. Based on the previous findings, the superiority of neural network-based models over tree-based models on tabular data is

under question; hence, the development of a robust treatment effect estimation model, which will exploit the advantages of both classes, is considered essential.

In this research, we propose an enhanced tree-based model for estimating the potential outcomes as well as the propensity score. The rationale of our approach is to exploit both the superiority of tree-based models for handling tabular data together with the notable property of neural network-based causal inference models in learning representations, which are used for estimating the outcomes for both the non-treatment and treatment cases. The proposed approach includes a loss function which is derived from non-parametric estimation theory leading to optimality of asymptotic characteristics [7, 12]. The effectiveness of the proposed model was compared against state-of-the-art tree-based and neural-network-based treatment effect estimation models on three collection of challenging causal inference datasets. The comprehensive numerical experiments provide empirical evidence about the effectiveness of the proposed model while statistical analysis further supports the effectiveness and robustness of the proposed approach. Summarizing, the main contributions of this research are:

- a new tree-based model is proposed, named C-XGBoost (treg), which extends our previous C-XGBoost model [13] by augmenting its architecture to simultaneously predict the propensity score alongside the two potential outcomes.
- an enhanced loss function for training the proposed model, which is equipped with a targeted regularization procedure based on non-parametric estimation theory, yielding asymptotically unbiased and efficient ATE estimates.
- we provide strong empirical and statistical evidence about the effectiveness the proposed model for treatment effect estimation.

The remainder of this paper is organized as follows: Section 2 presents a brief review on state-of-the-art models for treatment effect estimation. Section 3 presents a detailed description of the proposed model focusing on its advantages and main characteristics. Section 4 presents the description of the datasets. Section 5 presents a comprehensive experimental evaluation and statistical analysis. Finally, Section 6 provides a discussion of the proposed approach and the obtained results while Section 7 presents the conclusion and the main findings of this research.

2 Related work

Estimating causal effects is generally considered a complex and long-standing problem, which has recently received significant attention from the scientific community. Over the past decade, an increasing number of data-driven models were developed for causal effect estimation, exhibiting good performance and often promising results. Next, we present a brief overview of the most sophisticated and efficient tree-based and neural network-based models for estimating causal effects.

Kunzel et al. [14] introduced a new class of models for estimating treatment effects, where the primary aim is to predict the outcomes by including the treatment indicator along with the covariates, without giving the latter any special treatment. Specifically, the treatment indicator is included and processed by the base learner like any other feature. R-Forest is generally considered as the most effective model of this class, which utilizes the traditional Random-Forest [15] as a base learner. Nevertheless, the

considerable disadvantages of this approach are: (i) in case the treatment and control groups are very different in covariates, a single model is probably not sufficient to encode the different relevant dimensions and smoothness of features for both groups [16]; (ii) a tree-based base learner is not able to identify the significance of the treatment indicator as an input feature, which implies that it may completely ignore the treatment assignment during the training process[14].

Wager and Athey [17] extended the commonly used Random Forest (R-Forest) and proposed a new non-parametric ensemble tree-based model, named C-Forest for estimating heterogeneous treatment effects. The proposed model consists of a group of causal trees, which estimate the intervention effect at the leaves of the trees. A notable advantage of C-Forest is its robustness to the number of covariates, unlike other causal inference models as well as its ability to exhibit good performance even if the number of covariates is relatively small [17].

Shi et al. [7] introduced a neural network-based model, named DragonNet, for treatment effect estimation from observational data. The key innovations of DragonNet was (i) three-head architecture for predicting the treatment effects in both control and treatment groups and the propensity score (i.e., the likelihood that an instance will be assigned to the treatment group) (ii) a *targeted regularization technique* which leverages the sufficiency of the propensity score for the effective estimation of treatment effects. The experimental results revealed that DragonNet outperformed the state-of-the-art neural network-based models TARNet [6], GANITE [18] and CEVAE [19] in two collections of datasets. Furthermore, the authors highlighted the significance of targeted regularization in enhancing DragonNet's effectiveness, by showing that it is able to considerably improve DragonNet's estimation performance, building upon the sufficiency of the propensity score for causal adjustment.

In more recent works, Kiriakidou and Diou [8, 12] proposed a new approach for improving the estimation of heterogeneous and average treatment effects. In detail, the authors enriched the inputs of neural network-based treatment effect estimation models with the average of neighboring outcomes contained in the training dataset in both control and treatment groups. Along this line, the authors integrated the proposed methodology in the state-of-the-art models, DragonNet [7], NEDnet [7] and TARnet [6] and modified their architecture for including the average neighboring outcomes as inputs along with the covariates. The numerical experiments demonstrated the effectiveness of the proposed approach as well as the limitation of selecting the proper instances for the calculation of average of neighboring outcomes (number of neighboring instances and distance metric). Based on this work, in [20] the authors proposed an information-theoretic methodology, named Mutual Information-based Neighbor selection for Treatment effect estimation (MINT), for the a priori selection of the optimal number of neighboring instances in treatment and control group and distance metric with respect to pre-defined criteria. Finally, the authors conducted an experimental and statistical analysis for showcasing that the employment of the MINT methodology with the state-of-the-art DragonNet model is able to develop an accurate prediction model for treatment effect.

Kiriakidou et al. [13] proposed a new tree-based model, named C-XGBoost for the estimation of the treatment. The key idea behind their approach was to exploit the

superiority of tree-based models for handling tabular data together with the notable property of causal inference neural network-based models to learn representations that are useful for estimating the outcome for both the treatment and non-treatment cases. In addition, the authors proposed a new mean-squared-error-based loss function for training the proposed model. The numerical experiments were based on the prediction performance of C-XGBoost model was compared with the most widely used tree-based and neural-network based models on two collections of semi-synthetic datasets, which revealed the effectiveness of the proposed model.

In this research, we propose a tree-based model for estimating the potential outcomes as well as the propensity score. The motivation of the proposed approach is to exploit both the superiority of tree-based models for handling tabular data together with the notable property of neural network-based causal inference models in learning representations, which are used for estimating the outcomes for both the non-treatment and treatment cases. Moreover, a new loss function is proposed for training the proposed causal inference model, which is based on non-parametric estimation theory and consists on improving the model's ability to accurately estimate the treatment effects. It is worth mentioning that this work can be considered as an extension of our previous work [13] along two axes: (i) *architecture*: C-XGBoost (treg) adds a third output head for calculating the propensity score $\hat{g}(\mathbf{x}_i)$; (ii) *training objective*: the standard MSE-based loss is replaced by the targeted regularization-based loss, which provides asymptotic unbiasedness and efficiency guarantees. The numerical experiments and analysis provide empirical as well as statistical evidence about the effectiveness and robustness of the proposed approach.

3 C-XGBoost model with targeted regularization

Figure 1 depicts a high-level description of the proposed model's architecture, which takes as input an instance $\mathbf{x}_i$ and calculates the prediction of the conditional outcomes $\hat{Q}_0(\mathbf{x}_i;\theta)$ and $\hat{Q}_1(\mathbf{x}_i;\theta)$ for control and treatment groups, respectively as well as the prediction of the propensity score $\hat{g}(\mathbf{x}_i;\theta) = \sigma(z(\mathbf{x}_i;\theta))$ where θ is the vector with the model's parameters and σ is the sigmoid function.

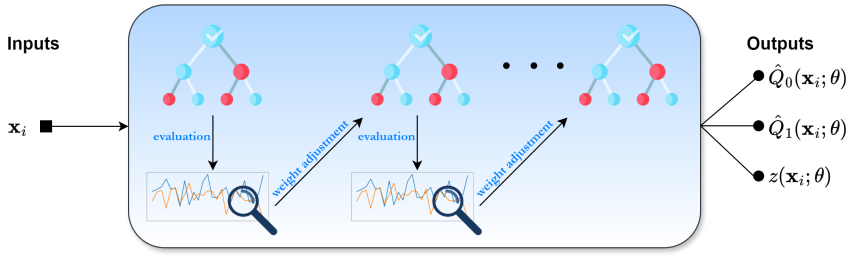


Fig. 1 High-level overview of the proposed model architecture and training procedure. Note that at each boosting iteration, a single multi-output tree is constructed with three values per leaf (for Q_0, Q_1, and z), rather than three separate trees.

3.1 Loss function

For training the proposed model, similar to [7, 13], we define the following loss function, which is the sum of a mean-square-error loss and cross-entropy loss, that is

$$\mathcal{L}(\mathbf{X}, t, y; \theta) = \frac{1}{n}\sum_{i}^{n}\Big[\mathcal{R}(\mathbf{x}_i, t_i, y_i; \theta) + \alpha\mathcal{C}(\hat{g}(\mathbf{x}_i), t_i)\Big] \tag{2}$$

where $\alpha > 0$ is a parameter, $\mathcal{C}$ is the cross-entropy loss [21], n is the dataset's (or batch) size and

$$\mathcal{R}(\mathbf{x}_i, t_i, y_i; \theta) = (1 - t_i)(y_i - Q_0(\mathbf{x}_i; \theta))^2 + t_i(y_i - Q_1(\mathbf{x}_i; \theta))^2 \tag{3}$$

Clearly, in case $\alpha = 0$, loss function (2) is reduced to the loss used in Kiriakidou et al. [13], since the cross-entropy term vanishes and the remaining mean-squared-error loss coincides with that of the baseline C-XGBoost model.

For further increasing the prediction ability of the proposed model, we employ *targeted regularization* [7] which consists of a modification to the objective function (2) equipped with a regularization term $r(\mathbf{x}_i, t_i, y_i; \theta, \epsilon)$, namely

$$\widetilde{\mathcal{L}}(\mathbf{X}, t, y; \theta, \epsilon) = \frac{1}{n}\sum_{i}^{n}\Big[\mathcal{R}(\mathbf{x}_i, t_i, y_i; \theta) + \alpha\mathcal{C}(\hat{g}(\mathbf{x}_i), t_i)\Big] + \beta\frac{1}{n}\sum_{i}^{n} r(\mathbf{x}_i, t_i, y_i; \theta, \epsilon) \tag{4}$$

where $\beta > 0$ is a parameter, $\mathcal{R}$ is a mean-square-error-based loss (3) and the regularization term is defined by

$$r(\mathbf{x}_i, t_i, y_i; \theta, \epsilon) = (1 - t_i)(y_i - \widetilde{Q}_0(\mathbf{x}_i; \theta))^2 + t_i(y_i - \widetilde{Q}_1(\mathbf{x}_i; \theta))^2$$

where

$$\begin{aligned}\widetilde{Q}_0(\mathbf{x}_i; \theta) &= Q_0(\mathbf{x}_i; \theta) - \epsilon\frac{1}{1 - \hat{g}(\mathbf{x}_i; \theta)}\\ \widetilde{Q}_1(\mathbf{x}_i; \theta) &= Q_1(\mathbf{x}_i; \theta) + \epsilon\frac{1}{\hat{g}(\mathbf{x}_i; \theta)}\end{aligned}$$

with ϵ to be a parameter. The motivation of the proposed loss function $\widetilde{\mathcal{L}}(\mathbf{X}, t, y; \theta, \epsilon)$ is based on non-parametric estimation theory and aims on enhancing the model's estimation ability of treatment effects.

Next, we present the following lemma for the estimator

$$\hat{\psi} = \frac{1}{n}\sum_{i}\hat{Q}_1^{\text{treg}}(\mathbf{x}_i) - \hat{Q}_0^{\text{treg}}(\mathbf{x}_i) \tag{5}$$

where $\hat{Q}_1^{\text{treg}}(\mathbf{x}_i) = \widetilde{Q}_1(\mathbf{x}_i; \theta)$ and $\hat{Q}_0^{\text{treg}}(\mathbf{x}_i) = \widetilde{Q}_0(\mathbf{x}_i; \theta)$, which states that the estimator $\hat{\psi}$ is asymptotically unbiased and efficient; namely, it converges to ψ at a fast rate

even when the nuisance parameters $\hat{Q}_0$, $\hat{Q}_1$, and $\hat{g}$ converge slowly, and it achieves the semiparametric efficiency bound by attaining the minimum asymptotic variance among all consistent estimators of ψ.

3.2 Theoretical properties

Before stating the main theoretical result, we explicitly state the standard causal inference assumptions required for its validity.

Assumption 1 (Unconfoundedness)**.** *Given the covariates $\mathbf{x}_i$, the treatment assignment T is independent of the potential outcomes, i.e., $\{Y^{(0)}, Y^{(1)}\} \perp T \mid X$.*

Assumption 2 (Positivity / Overlap)**.** *The propensity score is bounded away from 0 and 1, i.e., there exists $\eta \in (0, 0.5)$ such that $\eta \leq g(\mathbf{x}) \leq 1 - \eta$ for all $\mathbf{x} \in \mathcal{X}$.*

Assumption 3 (Consistency (SUTVA))**.** *The observed outcome satisfies $Y = T \cdot Y^{(1)} + (1 - T) \cdot Y^{(0)}$, with no interference between units and no hidden treatment versions.*

Assumption 4 (Regularity)**.** *The conditional outcome functions Q_0, Q_1 and the propensity score g are sufficiently smooth such that the estimators $\hat{Q}_0$, $\hat{Q}_1$ and $\hat{g}$ are consistent, i.e., converge to their true values in L_2.*

Lemma 1. *Under Assumptions 1–4, suppose that the tuple $(\hat{Q}_0, \hat{Q}_1, \hat{g}, \hat{\psi})$ satisfies the following non-parametric estimating equation*

$$\frac{1}{n} \sum_{i}^{n} \phi(y_i, t_i, \mathbf{x}_i; \hat{Q}_0, \hat{Q}_1, \hat{g}, \hat{\psi}) = 0 \tag{6}$$

where

$$\begin{aligned} \phi(y, t, x; Q_0, Q_1, g, \psi) = &\left(\frac{t}{g(x)} - \frac{1 - t}{1 - g(x)} \right) \Big((1 - t)(y - Q_0(t, x)) + t(y - Q_1(t, x)) \Big) \\ &+ Q_1(x) - Q_0(x) - \psi \end{aligned}$$

is the efficient influence curve of (1). Then, if C-XGBoost *(treg) is trained by minimizing*

$$\hat{\theta}, \hat{\epsilon} = \underset{\theta, \epsilon}{\operatorname{argmin}} \, \widetilde{\mathcal{L}}(\mathbf{X}, t, y; \theta, \epsilon),$$

the estimator $\hat{\psi}$ defined in Eq. (5) satisfies, asymptotically,

(i) $\hat{\psi}$ converges to ψ at a fast rate even if $\hat{Q}_0$, $\hat{Q}_1$ and $\hat{g}$ converge slowly.

(ii) $\hat{\psi}$ achieves the semiparametric efficiency bound, i.e., it has the lowest asymptotic variance among all consistent estimators of ψ.

Proof sketch. See Appendix A

The asymptotic guarantees of Lemma 1 rely on the consistency of $\hat{Q}_0$, $\hat{Q}_1$, and $\hat{g}$, which in practice is only approximated with a finite number of trees of limited depth. Nevertheless, the targeted regularization term enforces the non-parametric estimating equation (6) at every training iteration, providing a systematic bias-correction mechanism that remains active even in finite samples. Rather than relying solely on the

accuracy of the nuisance estimators, it directly steers $\hat{\psi}$ towards satisfying the efficiency condition, thereby mitigating the impact of finite-sample bias in $\hat{Q}_0$, $\hat{Q}_1$, and $\hat{g}$.

3.3 Training algorithm

At this point, we recall that the main aim of every algorithm dedicated to train a model of XGBoost-family is to optimize the following objective function which consists of the sum of training loss and regularization term.

$$\mathcal{J}(X, t, y; \theta) = \widetilde{\mathcal{L}}(X, t, y; \theta) + \sum_{k=1}^{K} \Omega(F_k),$$

where

$$\Omega(f_k) = \gamma_T T + \frac{1}{2}\|w\|^2,$$

is the regularization term that penalizes the model's complexity aiming to avoid overfitting, γ_T is the regularization parameter for the number of leaves, T is the number of leaves in each tree and w is the vector of scores (or weights) at the leaves. Along this line, Algorithm 1 summarizes the pseudocode for the development of the proposed model which takes as input the training dataset D, the number of boosting iterations K, the learning rate η, the regularization coefficient λ and the minimum loss reduction required for a split γ_T; while the output is the final ensemble model F_K.

Proposition 1 (Analytic gradients and Hessians of $\widetilde{\mathcal{L}}$)**.** *Let $\ell_i(\theta, \epsilon)$ denote the per-sample contribution to $\widetilde{\mathcal{L}}$. The first- and second-order partial derivatives with respect to the model outputs, evaluated at a sample $(\mathbf{x}_i, t_i, y_i)$, are given by:*

$$\frac{\partial \ell_i}{\partial Q_0(\mathbf{x}_i;\theta)} = -2(y_i - Q_0(\mathbf{x}_i;\theta)) - 2\beta(y_i - \widetilde{Q}_0(\mathbf{x}_i;\theta)), \qquad (t_i = 0), \tag{7}$$

$$\frac{\partial \ell_i}{\partial Q_1(\mathbf{x}_i;\theta)} = -2(y_i - Q_1(\mathbf{x}_i;\theta)) - 2\beta(y_i - \widetilde{Q}_1(\mathbf{x}_i;\theta)), \qquad (t_i = 1), \tag{8}$$

$$\frac{\partial \ell_i}{\partial z(\mathbf{x}_i;\theta)} = \alpha\big(\hat{g}(\mathbf{x}_i;\theta) - t_i\big) - 2\beta \begin{cases} \epsilon\,(y_i - \widetilde{Q}_0(\mathbf{x}_i;\theta))\,\dfrac{\hat{g}(\mathbf{x}_i;\theta)}{1-\hat{g}(\mathbf{x}_i;\theta)}, & t_i = 0, \\ \epsilon\,(y_i - \widetilde{Q}_1(\mathbf{x}_i;\theta))\,\dfrac{1-\hat{g}(\mathbf{x}_i;\theta)}{\hat{g}(\mathbf{x}_i;\theta)}, & t_i = 1, \end{cases} \tag{9}$$

and

$$\frac{\partial^2 \ell_i}{\partial Q_0(\mathbf{x}_i;\theta)^2} = 2(1+\beta), \qquad (t_i = 0), \tag{10}$$

$$\frac{\partial^2 \ell_i}{\partial Q_1(\mathbf{x}_i;\theta)^2} = 2(1+\beta), \qquad (t_i = 1), \tag{11}$$

$$\frac{\partial^2 \ell_i}{\partial z(\mathbf{x}_i;\theta)^2} = \alpha\, \hat{g}(\mathbf{x}_i;\theta)\big(1-\hat{g}(\mathbf{x}_i;\theta)\big) - 2\beta \begin{cases} \dfrac{\epsilon\,(y_i - \widetilde{Q}_0(\mathbf{x}_i;\theta))}{(1-\hat{g}(\mathbf{x}_i;\theta))^2}, & t_i = 0, \\ \dfrac{\epsilon\,(y_i - \widetilde{Q}_1(\mathbf{x}_i;\theta))}{\hat{g}(\mathbf{x}_i;\theta)^2}, & t_i = 1. \end{cases} \tag{12}$$

Proof. See Appendix B. □

Initially, the algorithm sets the ensemble model to a constant value. Then, for each boosting iteration, the algorithm computes the gradients and Hessians for each prediction (Steps 3-4), which are essential for understanding how much each prediction deviates from the true labels and the degree of curvature in the loss function, respectively. In Step 5, the computed gradients and Hessians are utilized to build a new decision tree T_k, whose construction involves selecting splits that maximize the gain, defined as the reduction in the loss function minus the regularization term $\Omega(F_k)$, which includes penalties for tree complexity based on regularization parameters λ and γ_T.

It is important to clarify that at each iteration k, the algorithm constructs a *single tree structure* T_k, not three separate trees. The multi-output capability is achieved by storing a vector of three values at each leaf node:

$$\mathbf{w}_j^{(k)} = [w_{Q_0,j}^{(k)}, w_{Q_1,j}^{(k)}, w_{z,j}^{(k)}]^\top$$

representing the contributions to the control outcome, treatment outcome, and propensity score predictions, respectively.

The tree construction process follows a greedy approach where at each node, potential splits are evaluated based on a composite gain metric that accounts for all three outputs simultaneously. For a candidate split that divides instances in region R into left (R_L) and right (R_R) child nodes, the gain is computed as:

$$\text{Gain} = \frac{1}{2}\left[\sum_{o\in\{Q_0,Q_1,z\}} \left(\frac{G_{o,L}^2}{H_{o,L}+\lambda} + \frac{G_{o,R}^2}{H_{o,R}+\lambda} - \frac{G_{o,R\cup L}^2}{H_{o,R\cup L}+\lambda}\right)\right] - \gamma_T \tag{13}$$

where $G_{o,L} = \sum_{i\in R_L} g_{o,i}^{(k)}$ and $H_{o,L} = \sum_{i\in R_L} h_{o,i}^{(k)}$ represent the sum of gradients and Hessians for output o in the left child, with similar definitions for the right child and parent node. The split that maximizes this composite gain is selected, ensuring that the tree structure is optimized for all prediction tasks simultaneously. This shared tree structure enforces a common representation across all outputs, promoting consistency in how the model interprets the covariate space, while being computationally more efficient than building three separate tree ensembles.

Next, in Step 6, for each leaf node j in the tree T_k, the weights $w_{Q_0,j}^{(k)}, w_{Q_1,j}^{(k)}$ and $w_{g,j}^{(k)}$ are calculated based on the gradients and Hessians associated with each leaf, which represent the contribution of each leaf node to the model's prediction. The model is then updated by adding a scaled contribution of the newly constructed tree T_k to the existing ensemble, where the scaling factor is determined by the learning rate η. The contribution is computed by summing the weights for each leaf node that a given

Algorithm 1 C-XGBoost Training with Targeted Regularization

Inputs:

D: Training dataset $\{\mathbf{x}_i, t_i, y_i\}_{i=1}^n$
K: Number of boosting iterations
η: Learning rate
λ: Regularization coefficient
γ: Minimum loss reduction required for a split

Output:

$F_K(x)$: Final ensemble model

1: Initialize the ensemble model $F_0(x)$ with:

$$Q_0^{(0)} = |\mathcal{I}_0|^{-1} \sum_{i \in \mathcal{I}_0} y_i, \quad Q_1^{(0)} = |\mathcal{I}_1|^{-1} \sum_{i \in \mathcal{I}_1} y_i, \quad z^{(0)} = \log(\bar{t}/(1-\bar{t})),$$

where $\mathcal{I}_0 = \{i : t_i = 0\}$, $\mathcal{I}_1 = \{i : t_i = 1\}$ and $\bar{t} = n^{-1} \sum_i t_i$.

2: **for** $k = 1$ to K **do**

3: Compute the gradients $g_{Q_0,i}^{(k)}$, $g_{Q_1,i}^{(k)}$ and $g_{z,i}^{(k)}$ for each prediction using (7), (8) and (9), respectively.

$$g_{Q_0,i}^{(k)} = \frac{\partial L}{\partial Q_0(x_i;\theta)} \quad g_{Q_1,i}^{(k)} = \frac{\partial L}{\partial Q_1(x_i;\theta)} \quad g_{g,i}^{(k)} = \frac{\partial L}{\partial g(x_i;\theta)}$$

4: Compute the Hessians $h_{Q_0Q_0,i}^{(k)}$, $h_{Q_1Q_1,i}^{(k)}$ and $h_{zz,i}^{(k)}$ for each prediction using (10), (11) and (12), respectively.

5: Build a tree T_k based on gradients and Hessians using procedure `TreeBuilding` that maximizes the gain from splitting, taking into account λ and γ.

6: For each leaf node j in tree T, compute the weights

$$w_{Q_0,j}^{(k)} = -\frac{\sum_{i \in R_j} g_{Q_0,i}^{(k)}}{\sum_{i \in R_j} h_{Q_0Q_0,i}^{(k)} + \lambda} \quad w_{Q_1,j}^{(k)} = -\frac{\sum_{i \in R_j} g_{Q_1,i}^{(k)}}{\sum_{i \in R_j} h_{Q_1Q_1,i}^{(k)} + \lambda} \quad w_{z,j}^{(k)} = -\frac{\sum_{i \in R_j} g_{z,i}^{(k)}}{\sum_{i \in R_j} h_{zz,i}^{(k)} + \lambda}$$

7: Update the model

$$F_k(x) = F_{k-1}(x) + \eta \sum_{j=1}^{J_k} \left(w_{Q_0,j}^{(k)} \mathbb{1}\{x \in R_j\} + w_{Q_1,j}^{(k)} \mathbb{1}\{x \in R_j\} + w_{z,j}^{(k)} \mathbb{1}\{x \in R_j\} \right)$$

where $\mathbb{1}$ is the indicator function and J_k is the total number of leaf nodes in the decision tree T_k.

8: **end for**

9: Return the final ensemble model $F_K(x)$.

input falls into. After, the iterative process is complete, the algorithm return the final

Algorithm 2 TreeBuilding

Inputs:

D_{node}: Instances at current node
$\{g^{(k)}_{Q_0,i}, g^{(k)}_{Q_1,i}, g^{(k)}_{z,i}\}$: Gradients for all instances
$\{h^{(k)}_{Q_0Q_0,i}, h^{(k)}_{Q_1Q_1,i}, h^{(k)}_{zz,i}\}$: Hessians for all instances
λ: Regularization coefficient
γ_T: Minimum loss reduction required for a split
depth: Current depth of the node
max-depth: Maximum allowed depth

Output:

T: Tree node (leaf or internal node)

1: **If** depth $\geq$ max-depth or $|D_{\text{node}}| <$ min_samples_split **then**
2: **Return** leaf with weights $w_o = -\sum_{i \in D_{\text{node}}} g^{(k)}_{o,i} / (\sum_{i \in D_{\text{node}}} h^{(k)}_{o,i} + \lambda)$ for each o
3: **End If**
4: best_gain $\leftarrow -\infty$, best_split $\leftarrow$ None
5: **For each** feature f and split value v:
6: Partition: $D_L \leftarrow \{i : x_{i,f} \leq v\}$, $D_R \leftarrow \{i : x_{i,f} > v\}$
7: Compute $G_{o,L} = \sum_{i \in D_L} g^{(k)}_{o,i}$, $H_{o,L} = \sum_{i \in D_L} h^{(k)}_{o,i}$ for each o (similarly for R and parent P)
8: Calculate gain using

$$\text{Gain} = \frac{1}{2} \sum_o \left(\frac{G^2_{o,L}}{H_{o,L} + \lambda} + \frac{G^2_{o,R}}{H_{o,R} + \lambda} - \frac{G^2_{o,P}}{H_{o,P} + \lambda} \right) - \gamma_T$$

9: **If** Gain $>$ best_gain: best_gain $\leftarrow$ Gain, best_split $\leftarrow (f, v)$
10: **End For**
11: **If** best_gain ≤ 0: **Return** leaf as in Step 2
12: Split D_{node} using best_split into D_L and D_R
13: $T_{\text{left}} \leftarrow$ TreeBuilding(D_L, gradients, Hessians, λ, γ_T, depth+1, max-depth)
14: $T_{\text{right}} \leftarrow$ TreeBuilding(D_R, gradients, Hessians, λ, γ_T, depth+1, max-depth)
15: **Return** internal node with best_split, T_{left}, T_{right}

ensemble model F_K, which is able to provide for a given sample $\mathbf{x}_i$ estimation of the conditional outcomes, $Q_0(\mathbf{x}_i; \theta)$ and $Q_1(\mathbf{x}_i; \theta)$, as well as the propensity score $\hat{g}(\mathbf{x}_i; \theta)$.

Note: It is worth mentioning that Algorithm 1 constructs a *single tree structure* at each iteration k, not three separate trees. The multi-output capability is achieved through the XGBoost's `multi_strategy`="`multi_output_tree`" parameter, which enables each leaf node to store a vector of outputs rather than a scalar value. This is fundamentally different from building three independent tree ensembles. The splitting decisions are made by jointly considering the gradients and Hessians from all

three outputs (Eq. (13)), ensuring that the tree structure is optimized for all prediction tasks simultaneously. After K iterations, the final model consists of K trees (not $3K$), where each tree contributes to all three predictions.

4 Data

In general, the evaluation of the effectiveness and robustness of causal inference models is regarded as a considerable challenging task due to the inherent absence of the counterfactual outcomes [22]. For mitigating this issue, we employed three collections of semi-synthetic datasets:

- **Synthetic** consists of a collection of toy causal-inference datasets, which was introduced by Louizos et al. [19]. The generation of each dataset is using based on the following process:

$$w_i \sim \text{Bern}(0.5), \tag{14}$$
$$t_i \,|\, w_i \sim \text{Bern}(0.75w_i + 0.25(1 - w_i)), \tag{15}$$
$$\mathbf{x}_i \,|\, w_i \sim \mathcal{N}(w_i, \sigma_{z_1}^2 w_i + \sigma_{z_0}^2(1 - w_i)), \tag{16}$$
$$y_i \,|\, t_i, w_i \sim \text{Bern}(f(3(w_i + 2(2t_i - 1)))), \tag{17}$$

  where W is the hidden confounder variable, f is the logistic sigmoid function, $\sigma_{z_0} = 3$ and $\sigma_{z_1} = 5$ are parameters, while the treatment t_i and the proxy variable $\mathbf{x}_i$ consist of a mixture of Bernoulli and Gaussian distributions, respectively. Each dataset contains 5000 samples and 1000 covariates, while the collection consists of 100 datasets, partitioned using an 80/20 scheme into training (4000 samples) and test (1000 samples) sets as in [8, 19].
- **IHDP** consists of a collection of datasets, which was provided from the Infant Health and Development Program [23]. Each dataset consists of 748 instances (609 in the control group, 139 in the treatment group) with 25 covariates, partitioned using an 80/20 scheme into training (598 samples) and test (150 samples) sets. The "A" collection of the NPCI package [24] is used, consisting of 1000 datasets, and results are reported as averages over all 1000 datasets.
- **ACIC** collection of semi-synthetic datasets was original developed for the 2018 Atlantic Causal Inference Conference competition data [25]. Each dataset of this collection was obtained from the linked birth and infant death data and constitutes a sample from a distinct distribution [26]. The instances in each dataset are generated through an iterative process with different treatment selection and outcome functions. Similar to [7, 12], 5 datasets of size 5000 and 11 datasets of size 10000 were randomly selected, each partitioned using an 80/20 scheme into training and test sets (4000/1000 and 8000/2000 samples, respectively).

5 Experimental analysis

In this section, we focus on providing strong empirical and statistical evidence about the efficiency of the proposed model. Next, we evaluate the performance of:

- "R-Forest", which follows S-learner approach [14] employing Random-Forest [15] as base model.
- "C-Forest", which stands for C-Forest model [17].
- "DragonNet", which stands for DragonNet model [7].
- "MINT-DragonNet", which stands for MINT-DragonNet [20].
- "C-XGBoost", which stands for the proposed C-XGBoost model [13].
- "C-XGBoost (treg)", which stands for the proposed C-XGBoost model enhanced with targeted regularization.[1]

The causal inference models R-Forest, C-Forest, DragonNet and MINT-DragonNet were used with their original optimized settings and hyper-parameters while both C-XGBoost and C-XGBoost (treg) were employed using the following hyperparameters: $K = 200$, $\lambda = 1.0$, $\gamma_T = 0.5$, max-depth $= 4$, and tree-method $=$ "hist" while the rest parameters were set as default. Notice that the composite gain in Eq. (13) and the multi-output leaf structure are *natively supported* by XGBoost ≥ 2.0 via the parameter `multi_strategy=''multi_output_tree''` combined with a custom objective that returns per-sample gradient/Hessian vectors of length 3; the full derivation of the latter is provided in the Appendix B.

The performance of all causal inference models is evaluated utilizing the performance metrics: absolute error in ATE

$$|\epsilon_{ATE}| = \left| \frac{1}{n} \sum_{i=1}^{n} (y(\mathbf{x}_i, 1) - y(\mathbf{x}_i, 0)) - \frac{1}{n} \sum_{i=1}^{n} (Q_1(\mathbf{x}_i) - Q_0(\mathbf{x}_i)) \right|,$$

and expected Precision in Estimation of Heterogeneous Effect

$$\text{PEHE} = \frac{1}{n} \sum_{i=1}^{n} \Big((y(\mathbf{x}_i, 1) - y(\mathbf{x}_i, 0)) - (Q_1(\mathbf{x}_i) - Q_0(\mathbf{x}_i)) \Big)^2.$$

Notice that ϵ_{ATE} and PEHE compare the models as estimators (ATE within the same dataset) and as predictors (ATE across datasets), respectively. It is worth mentioning that these two metrics were selected since they constitute the most widely utilized metrics for comparing the performance of models for estimating treatment effects [7, 9, 19].

The distributions across datasets within each collection are inherently different; therefore, direct quantitative comparisons of model performance across different datasets would be misleading. For this purpose, we employ a qualitative approach through statistical ranking methods that account for these distributional differences. In detail, for mitigating the influence of specific datasets on the evaluation process for each collection, a statistical analysis is conducted for evaluating the hypothesis H_0 that all causal inference models exhibit equally accurate predictions. To this end, non-parametric Friedman Aligned-Ranks (FAR) test [27] and the Li post-hoc test [28] with significance level $\alpha = 5\%$ are employed. The former is employed for ranking the causal inference models based on a performance metric while the latter is used for

[1] The implementation code of both C-XGBoost and C-XGBoost (treg) can be found in https://github.com/ioannislivieris/C-XGBoost

examining the existence of considerable differences and variations in their predictions, without any assumption about the distribution of the performance scores [20, 29].

5.1 Numerical experiments

Table 1 summarizes the statistical comparison of the evaluated causal inference models for Synthetic collection of datasets, relative to both performance metrics. C-XGBoost (treg) exhibits the top performance, demonstrating the best FAR for both ϵ_{ATE} and PEHE metrics. Furthermore, the interpretation of Li post-hoc test suggests that there are considerable differences between the performance of the proposed model and the traditional tree-based and neural-network-based models, which indicates the superiority of our approach. Therefore, we are able to deduce that the proposed C-XGBoost (treg) model is a better estimator as well as a better predictor than the traditional causal inference models.

Table 2 presents the statistical comparison of all evaluated models for IHDP collection of datasets based on $|\epsilon_{ATE}|$ and $PEHE$. C-XGBoost (treg) and MINT-DragonNet report the best performance relative to both performance metrics while the traditional tree-based model report the worst performance. By taking into consideration that datasets' characterics of this collection, we are able to deduce that the traditional tree-based models C-Forest, R-Forest and C-XGBoost were affected by the imbalanced in the datasets while the proposed C-XGBoost (treg) present competitive and frequently superior performance to neural-network-based models. In addition, from the comparison of C-XGBoost (treg) and C-XGBoost, we conclude that targeted regularization had a considerable positive effect to the performance of the proposed model.

Table 3 presents the statistical comparison of neural-network-based and tree-based causal inference models for ACIC collection of datasets, relative to both performance metrics. The proposed C-XGBoost (treg) exhibits top performance as regards both metrics, following by the tree-based models R-Forest and C-XGBoost. Moreover, neural-network-based and C-Forest report poor performance for this collection of datasets. Furthermore, the proposed C-XGBoost (treg) exhibits better (lower) FAR score than C-XGBoost, which implies targeted regularization increased the proposed model's prediction ability.

Models	FAR	Li post-hoc test	
		p-value	H_0
C-XGBoost (treg)	126.57	-	-
C-XGBoost	178.24	0.0350	Reject
R-Forest	187.23	0.0136	Reject
MINT-DragonNet	405.10	0.0	Reject
DragonNet	446.09	0.0	Reject
C-Forest	459.77	0.0	Reject

(a) ϵ_{ATE}

Models	FAR	Li post-hoc test	
		p-value	H_0
C-XGBoost (treg)	169.50	-	-
C-XGBoost	219.76	0.0040	Reject
MINT-DragonNet	263.22	0.0	Reject
DragonNet	293.12	0.0	Reject
C-Forest	307.08	0.0	Reject
R-Forest	550.31	0.0	Reject

(b) PEHE

Table 1 FAR test and Li post-hoc test for Synthetic collection of datasets based on (a) ϵ_{ATE} and (b) PEHE

Models	FAR	Li post-hoc test	
		p-value	H_0
C-XGBoost (treg)	234.26	-	-
MINT-DragonNet	248.43	0.5632	Fail
C-Forest	250.08	0.5429	Fail
DragonNet	256.79	0.4505	Fail
R-Forest	269.19	0.2609	Fail
C-XGBoost	544.25	0.0	Reject

(a) ϵ_{ATE}

Models	FAR	Li post-hoc test	
		p-value	H_0
C-XGBoost (treg)	203.96	-	-
MINT-DragonNet	208.32	0.8588	Fail
DragonNet	208.46	0.8582	Fail
R-Forest	240.94	0.4821	Fail
C-XGBoost	439.83	0.0	Reject
C-Forest	501.49	0.0	Reject

(b) PEHE

Table 2 FAR test and Li post-hoc test for IHDP collection of datasets based on (a) ϵ_{ATE} and (b) PEHE

Models	FAR	Li post-hoc test	
		p-value	H_0
C-XGBoost (treg)	34.05	-	-
R-Forest	35.47	0.8893	Fail
C-XGBoost	50.29	0.4977	Fail
MINT-DragonNet	60.82	0.0702	Fail
DragonNet	63.52	0.0322	Reject
C-Forest	64.82	0.0215	Reject

(a) ϵ_{ATE}

Models	FAR	Li post-hoc test	
		p-value	H_0
C-XGBoost (treg)	29.52	-	-
C-XGBoost	35.88	0.5313	Fail
R-Forest	42.52	0.2993	Fail
MINT-DragonNet	64.76	0.0011	Reject
DragonNet	66.47	0.0005	Reject
C-Forest	69.82	0.0001	Reject

(b) PEHE

Table 3 FAR test and Li post-hoc test for ACIC collection of datasets based on (a) ϵ_{ATE} and (b) PEHE

Summarizing, it is worth highlighting the C-XGBoost (treg) presented the best overall performance in all three collections of datasets. By talking into consideration that the selected collections possess different characteristics and properties, we conclude that the proposed model constitutes a promising alternative to traditional models as both an estimator and prediction.

5.2 Ablation study

Next, an ablation study is conducted for evaluating and measuring the performance of the proposed C-XGBoost (treg) model using various values of parameter β in order to study the effectiveness of targeted regularization to the model's performance. Notice that in case $\beta = 0$ then the proposed loss function (3) is reduced to (2), which implies that no targeted regularization is employed.

As regards IHDP, higher values of β notably enhance model performance while for low or zero values of β C-XGBoost (treg) model exhibits poorer outcomes. This implies the importance of regularization in mitigating the imbalance present in this collection of datasets. The statistical analysis for Synthetic collection of datasets indicates that moderate values of β, particularly around 0.4 and 0.6, present the best results, with performance deteriorating under excessive or minimal regularization. Finally, for the ACIC collection of datasets, targeted regularization proves crucial, with optimal performance observed at higher β values (0.8 and 1.0), underscoring its role in managing complex data.

C-XGBoost (treg)	FAR	Li post-hoc test	
		p-value	H_0
$\beta = 1.0$	247.23	-	-
$\beta = 0.8$	250.42	0.8964	Fail
$\beta = 0.6$	250.82	0.8951	Fail
$\beta = 0.4$	251.93	0.8911	Fail
$\beta = 0.2$	253.55	0.8849	Fail
$\beta = 0.0$	549.05	0.0	Reject

(a) ϵ_{ATE}

C-XGBoost (treg)	FAR	Li post-hoc test	
		p-value	H_0
$\beta = 1.0$	247.53	-	-
$\beta = 0.8$	251.75	0.8633	Fail
$\beta = 0.4$	254.91	0.8481	Fail
$\beta = 0.2$	255.48	0.8451	Fail
$\beta = 0.6$	263.12	0.7933	Fail
$\beta = 0.0$	430.21	0.0	Reject

(b) PEHE

Table 4 FAR test and Li post-hoc test for IHDP collection of datasets for various values of parameter β based on (a) ϵ_{ATE} and (b) PEHE

C-XGBoost (treg)	FAR	Li post-hoc test	
		p-value	H_0
$\beta = 0.6$	278.42	-	-
$\beta = 1.0$	284.05	0.8185	Fail
$\beta = 0.8$	288.92	0.7865	Fail
$\beta = 0.0$	299.81	0.6785	Fail
$\beta = 0.4$	320.62	0.3194	Fail
$\beta = 0.2$	331.17	0.1476	Fail

(a) ϵ_{ATE}

C-XGBoost (treg)	FAR	Li post-hoc test	
		p-value	H_0
$\beta = 0.4$	205.91	-	-
$\beta = 0.6$	239.57	0.1698	Fail
$\beta = 0.8$	285.42	0.0014	Reject
$\beta = 1.0$	332.25	0.0	Reject
$\beta = 0.2$	360.46	0.0	Reject
$\beta = 0.0$	379.38	0.0	Reject

(b) PEHE

Table 5 FAR test and Li post-hoc test for Synthetic collection of datasets for various values of parameter β based on (a) ϵ_{ATE} and (b) PEHE

C-XGBoost (treg)	FAR	Li post-hoc test	
		p-value	H_0
$\beta = 0.8$	38.70	-	-
$\beta = 1.0$	45.88	0.4794	Fail
$\beta = 0.6$	47.01	0.4428	Fail
$\beta = 0.4$	53.70	0.2112	Fail
$\beta = 0.0$	59.35	0.0451	Reject
$\beta = 0.2$	64.35	0.0216	Reject

(a) ϵ_{ATE}

C-XGBoost (treg)	FAR	Li post-hoc test	
		p-value	H_0
$\beta = 1.0$	41.35	-	-
$\beta = 0.8$	43.11	0.8619	Fail
$\beta = 0.6$	47.11	0.8050	Fail
$\beta = 0.4$	57.01	0.4714	Fail
$\beta = 0.0$	58.64	0.3903	Fail
$\beta = 0.2$	61.76	0.2429	Fail

(b) PEHE

Table 6 FAR test and Li post-hoc test for ACIC collection of datasets for various values of parameter β based on (a) ϵ_{ATE} and (b) PEHE

Summarizing, the statistical analysis confirm that targeted regularization improved the proposed model's performance while its optimal value is context-dependent and varies with dataset complexity. The experiments suggest that for challenging and complex dataset a value around 1.0 for parameter β is recommended, while for less demanding datasets moderate values of β around 0.5 are preferable.

6 Discussion

In this section, we provide an in-depth discussion of the proposed approach of this work along with a comprehensive analysis of the experimental results. Through this discussion, the primary aim is to highlight the strengths, limitations, and potential applications of the proposed method, as well as its contribution to advancing the field.

In this research, we proposed an extended tree-based model, named C-XGBoost (treg), which builds upon our previous work [13] by augmenting the architecture with propensity score estimation and an enhanced loss function equipped with targeted regularization. The motivation of the proposed approach is to exploit both the superiority of tree-based models for handling tabular data along with the notable property of neural network-based causal inference models in learning representations for estimating the outcomes for both the non-treatment and treatment use cases. A significant advantage of the proposed model is that it is trained using a new loss function, which is based on non-parametric estimation theory and consists in improving the model's ability to accurately estimate the treatment effects. In detail, the loss function is equipped with targeted regularization, which constitutes an efficient regularization procedure aiming at effectively estimate the treatment effects through leveraging the propensity score. Additionally, the proposed model inherits the advantages of original XGBoost model including advanced tree pruning ("depth-wise") where trees are split in a greedy manner, regularization techniques for avoiding overfitting/bias as well as efficiently handling missing data [30].

The effectiveness and robustness of the introduced model was examined by conducting a series of experiments on three widely used collections of datasets (IHDP, Synthetic and ACIC), which focused on evaluating the performance of the proposed C-XGBoost (treg) model against that of the most efficient tree-based (C-Forest and R-Forest) and neural network-based (DragonNet and MINT-DragonNet) in the area. The analysis of the experiments was based on FAR test and Li post-hoc test, which were employed for ranking the causal inference models based on a performance metric and for examining the existence of considerable differences in their predictions, respectively.

The ablation study on the C-XGBoost (treg) model provides insights regarding the impact of targeted regularization on model's performance for various values of the regularization parameter β. For the imbalanced IHDP collection of datasets as well as for the most complex collection ACIC, the statistical analysis demonstrated that higher values (0.8 and 1.0) of the regularization parameter β improve the model's performance relative to both performance metrics. In contrast, for low values of parameters β or in case β is set to zero (no targeted regularization), the model exhibited substantially poorer performance. Hence, we are able to conclude that targeted regularization assists the model handle the inherent difficulties of both collection of datasets. The analysis based on the impact of parameter β on the performance of C-XGBoost (treg) model relative to the Synthetic collection of datasets revealed that the proposed model benefits from moderate values of β, with $\beta = 0.4$ showing the best results. Nonetheless, both metrics exhibit a trend where excessive or minimal regularization degrades performance, suggesting the need for careful tuning of β. These findings highlight that (i) while targeted regularization can enhance model performance, its optimal level may vary depending on the dataset's complexity and inherent properties (ii) Synthetic dataset's characteristics may not require as stringent regularization compared to IHDP and ACIC, which are more complex.

Based on the previous discussion and findings, we are able to conclude that targeted regularization had a considerable positive effect to the performance of the proposed

model. In addition, it is worth mentioning that while it is able to considerably enhance C-XGBoost's performance, its optimal level may vary depending on the dataset's complexity and inherent properties. Our experiments on three collection of datasets suggest that values between 0.8 and 1.0 constitute good choices; nevertheless, this is something which should be further studied in future research.

It is worth noting that the evaluation was conducted exclusively on semi-synthetic datasets, which, although standard practice in the causal inference literature since counterfactual outcomes are never observed in real data, often exhibit simpler confounding structures and known propensity scores by construction. In real-world settings, confounding may be substantially more complex, overlap assumptions considerably harder to verify, and the ground truth ATE unknowable without auxiliary evidence such as sensitivity analyses or domain knowledge. The evaluation of C-XGBoost (treg) on real-world benchmarks, where advanced machine learning models have demonstrated strong effectiveness in capturing complex nonlinear patterns in observational data [31–34] and where treatment effect estimation is of direct practical relevance; therefore, constitutes a primary direction of future work.

Finally, the conducted research aimed at presenting a new tree-based model for estimating the treatment effects as well as the propensity score. The numerical experiments and analysis provide empirical as well as statistical evidence about the efficiency and robustness of the proposed approach and its ability to be applied to causal inference problems without the consideration of any additional modifications and/or constraints. Although the current work focuses on point estimation, uncertainty quantification is a critical requirement in safety-critical applications of causal inference. Extending C-XGBoost (treg) to produce calibrated uncertainty estimates, via conformal prediction intervals, quantile regression objectives within the XGBoost framework, or Bayesian ensemble methods, constitutes a natural and compelling next step in this research.

7 Conclusions

The main contribution of this research is the development of C-XGBoost (treg) model for the prediction of potential outcomes as well as the propensity score. The advantages of the proposed model are (i) exploitation of the strong prediction abilities of XGBoost algorithm (ii) the remarkable property of causal inference neural network-based models to learn representations for estimating the outcome for both the treatment and non-treatment cases. In addition, the proposed model is trained with an enhanced loss function, which augments that proposed in [13] with a targeted regularization procedure, which favors models that inherently possess non-parametric optimality in their asymptotic characteristics. The performance of the proposed C-XGBoost (treg) model was evaluated against state-of-the-art treatment estimation models on three semi-synthetic challenging collections of datasets. The experimental analysis demonstrated that the proposed model exhibited the top performance both as an estimator and as predictor. In addition, statistical analysis, provided strong evidence about the effectiveness and robustness of the proposed approach, theoretical securing its efficiency in treatment effect estimation.

A limitation of this work can be considered the influence of hyperparameter settings to the effectiveness of proposed model. In our preliminary (unreported) experiments, we identified that C-XGBoost with or without targeted regularization is sensitive to different configurations while some hyperparameters, such as K, may significantly impact its effectiveness. Nevertheless, this is no surprise for a model from the XGBoost-family; hence, a systematic and efficient tuning procedure is considered a priority in our research tasks. It is worth mentioning that addressing this limitation through experimentation not only aims to increase the performance of the proposed model but also contributes to the understanding of the model's behavior. Another limitation of the proposed work is the fact that the evaluation process was performed on semi-synthetic collections of datasets. Therefore, the application and employment of C-XGBoost (treg) model on real-world benchmarks is definitely included in our future work. This will allow to assess the proposed model's performance in more dynamic environments, providing a more accurate representation of its effectiveness in practical settings as well as enable us to identify any potential limitations or areas for further refinement, which may not have surfaced during the initial evaluation.

Our future work is concentrated on enriching the inputs of C-XGBoost (treg) model by employing MINT methodology for leveraging information from the outcome of neighboring instances. This scope of integrating this information-theoretic methodology is further improve the proposed model's ability for predicting treatment estimation and reduce the bias by enriching the covariate space. Another interesting approach for future research is the utilization of data-splitting in treatment effect estimation [35]. Although this approach was not found effective for neural network-based models [7] it is not clear that it is also true for tree-based ensemble models as well.

Declarations

Availability of data and material: Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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Compliance with ethical standards.

A Proof of Lemma 1

Proof sketch. Under Assumptions 1–4, the efficient influence curve of the ATE functional (1) is given by ϕ as defined above [35, 36]. Any regular asymptotically

linear estimator of ψ must satisfy the estimating equation (6), and the semiparametric efficiency bound is achieved by the estimator whose influence curve equals ϕ. By differentiating $\widetilde{\mathcal{L}}$ with respect to ϵ and setting it to zero, we obtain

$$0 = \partial_\epsilon \widetilde{\mathcal{L}}(\mathbf{X}, t, y; \theta, \epsilon)\big|_{\hat{\epsilon}} = \frac{\beta}{n} \sum_{i=1}^{n} \partial_\epsilon \, r(\mathbf{x}_i, t_i, y_i; \theta, \epsilon)\big|_{\hat{\epsilon}},$$

which, after direct calculation, reduces to

$$\frac{1}{n} \sum_{i=1}^{n} \phi(y_i, t_i, \mathbf{x}_i; \hat{Q}_0, \hat{Q}_1, \hat{g}, \hat{\psi}) = 0,$$

thereby forcing $(\hat{Q}_0, \hat{Q}_1, \hat{g}, \hat{\psi})$ to satisfy (6). Properties (i) and (ii) then follow from standard semiparametric theory [35, 36]. Finally, the double robustness of $\hat{\psi}$, namely its consistency when either $(\hat{Q}_0, \hat{Q}_1)$ or $\hat{g}$ is correctly specified, follows as an additional consequence of satisfying (6). □

B Proof of Proposition 1

We derive the first- and second-order partial derivatives of $\widetilde{\mathcal{L}}$ with respect to each model output. We work with the per-sample contribution to $\widetilde{\mathcal{L}}$ and adopt the shorthand

$$Q_k \coloneqq Q_k(\mathbf{x}_i; \theta), \quad \widetilde{Q}_k \coloneqq \widetilde{Q}_k(\mathbf{x}_i; \theta), \quad \hat{g} \coloneqq \hat{g}(\mathbf{x}_i; \theta), \quad z \coloneqq z(\mathbf{x}_i; \theta),$$

for $k \in \{0, 1\}$. The per-sample loss reads

$$\begin{aligned} \ell_i(\theta, \epsilon) = (1 - t_i)(y_i - Q_0)^2 + t_i(y_i - Q_1)^2 + \alpha \, \mathcal{C}(\hat{g}, t_i) \\ + \beta \Big[(1 - t_i)(y_i - \widetilde{Q}_0)^2 + t_i(y_i - \widetilde{Q}_1)^2 \Big], \end{aligned} \tag{18}$$

where $\hat{g} = \sigma(z)$, σ denotes the logistic sigmoid,

$$\widetilde{Q}_0 = Q_0 - \frac{\epsilon}{1 - \hat{g}}, \qquad \widetilde{Q}_1 = Q_1 + \frac{\epsilon}{\hat{g}},$$

and $\mathcal{C}(\hat{g}, t_i) = -t_i \log \hat{g} - (1 - t_i) \log(1 - \hat{g})$.

B.1 Derivatives with respect to Q_0

The terms in ℓ_i depending on Q_0 are

$$(1 - t_i)(y_i - Q_0)^2 + \beta(1 - t_i)(y_i - \widetilde{Q}_0)^2.$$

Both terms vanish identically for $t_i = 1$; hence Q_0 receives a gradient signal only from control units ($t_i = 0$), and all subsequent derivatives are stated for $t_i = 0$. Since

$\widetilde{Q}_0 = Q_0 - \epsilon/(1-\hat{g})$, we have $\partial\widetilde{Q}_0/\partial Q_0 = 1$, so the chain rule gives

$$\frac{\partial}{\partial Q_0}\big[(y_i - Q_0)^2\big] = -2(y_i - Q_0),$$
$$\frac{\partial}{\partial Q_0}\Big[\beta(y_i - \widetilde{Q}_0)^2\Big] = -2\beta(y_i - \widetilde{Q}_0).$$

Summing,

$$\frac{\partial \ell_i}{\partial Q_0} = -2(y_i - Q_0) - 2\beta(y_i - \widetilde{Q}_0). \tag{19}$$

Differentiating once more,

$$\frac{\partial^2}{\partial Q_0^2}\big[(y_i - Q_0)^2\big] = 2,$$
$$\frac{\partial^2}{\partial Q_0^2}\Big[\beta(y_i - \widetilde{Q}_0)^2\Big] = 2\beta\left(\frac{\partial\widetilde{Q}_0}{\partial Q_0}\right)^2 = 2\beta,$$

and summing,

$$\frac{\partial^2 \ell_i}{\partial Q_0^2} = 2(1+\beta). \tag{20}$$

B.2 Derivatives with respect to Q_1

The terms in ℓ_i depending on Q_1 are

$$t_i(y_i - Q_1)^2 + \beta\, t_i(y_i - \widetilde{Q}_1)^2.$$

Both terms vanish identically for $t_i = 0$; hence Q_1 receives a gradient signal only from treated units ($t_i = 1$), and all subsequent derivatives are stated for $t_i = 1$. Since $\widetilde{Q}_1 = Q_1 + \epsilon/\hat{g}$, we have $\partial\widetilde{Q}_1/\partial Q_1 = 1$, so the chain rule gives

$$\frac{\partial}{\partial Q_1}\big[(y_i - Q_1)^2\big] = -2(y_i - Q_1),$$
$$\frac{\partial}{\partial Q_1}\Big[\beta\, t_i(y_i - \widetilde{Q}_1)^2\Big] = -2\beta(y_i - \widetilde{Q}_1).$$

Summing,

$$\frac{\partial \ell_i}{\partial Q_1} = -2(y_i - Q_1) - 2\beta(y_i - \widetilde{Q}_1). \tag{21}$$

Differentiating once more,

$$\frac{\partial^2}{\partial Q_1^2}\left[(y_i - Q_1)^2\right] = 2,$$
$$\frac{\partial^2}{\partial Q_1^2}\left[\beta(y_i - \widetilde{Q}_1)^2\right] = 2\beta\left(\frac{\partial \widetilde{Q}_1}{\partial Q_1}\right)^2 = 2\beta,$$

and summing,

$$\frac{\partial^2 \ell_i}{\partial Q_1^2} = 2(1+\beta). \tag{22}$$

A.3 Derivatives with respect to z

Since $\hat{g} = \sigma(z)$, we have

$$\frac{\partial \hat{g}}{\partial z} = \hat{g}(1-\hat{g}).$$

The loss ℓ_i depends on z through three terms, which we differentiate in turn.

(i) Cross-entropy. Expanding the definition of $\mathcal{C}(\hat{g}, t_i)$,

$$\begin{aligned}\frac{\partial \mathcal{C}}{\partial z} &= \frac{\partial}{\partial z}[-t_i \log \hat{g} - (1-t_i)\log(1-\hat{g})] \\ &= -\frac{t_i}{\hat{g}} \cdot \hat{g}(1-\hat{g}) + \frac{1-t_i}{1-\hat{g}} \cdot \hat{g}(1-\hat{g}) \\ &= -t_i(1-\hat{g}) + (1-t_i)\hat{g} \\ &= \hat{g} - t_i.\end{aligned}$$

Differentiating once more,

$$\frac{\partial^2 \mathcal{C}}{\partial z^2} = \frac{\partial \hat{g}}{\partial z} = \hat{g}(1-\hat{g}).$$

(ii) Targeted regularization, $t_i = 0$. From $\widetilde{Q}_0 = Q_0 - \epsilon/(1-\hat{g})$, the chain rule through $\hat{g}$ gives

$$\frac{\partial \widetilde{Q}_0}{\partial z} = \frac{\partial}{\partial \hat{g}}\left[-\frac{\epsilon}{1-\hat{g}}\right] \cdot \frac{\partial \hat{g}}{\partial z} = -\frac{\epsilon}{(1-\hat{g})^2} \cdot \hat{g}(1-\hat{g}) = -\frac{\epsilon\, \hat{g}}{1-\hat{g}}.$$

Setting $r_i^{(0)} \coloneqq y_i - \widetilde{Q}_0$ and applying the chain rule,

$$\frac{\partial}{\partial z}\left[\beta(y_i - \widetilde{Q}_0)^2\right] = -2\beta\, r_i^{(0)} \cdot \frac{\partial \widetilde{Q}_0}{\partial z} = 2\beta\, r_i^{(0)} \cdot \frac{\epsilon\, \hat{g}}{1-\hat{g}}.$$

Differentiating once more via the product rule,

$$\frac{\partial^2}{\partial z^2}\Big[\beta(y_i-\widetilde{Q}_0)^2\Big] = 2\beta\left[\frac{\partial r_i^{(0)}}{\partial z}\cdot\frac{\epsilon\,\hat{g}}{1-\hat{g}} + r_i^{(0)}\cdot\frac{\partial}{\partial z}\left(\frac{\epsilon\,\hat{g}}{1-\hat{g}}\right)\right].$$

Since $\partial r_i^{(0)}/\partial z = -\partial\widetilde{Q}_0/\partial z = \epsilon\hat{g}/(1-\hat{g})$, the first term equals $\epsilon^2\hat{g}^2/(1-\hat{g})^2 \geq 0$. The second term is weighted by the residual $r_i^{(0)}$, which vanishes at convergence; dropping it yields

$$\frac{\partial^2}{\partial z^2}\Big[\beta(y_i-\widetilde{Q}_0)^2\Big] \approx 2\beta\cdot\frac{\epsilon(y_i-\widetilde{Q}_0)}{(1-\hat{g})^2}. \tag{23}$$

(iii) Targeted regularization, $t_i = 1$. From $\widetilde{Q}_1 = Q_1 + \epsilon/\hat{g}$, the chain rule through $\hat{g}$ gives

$$\frac{\partial\widetilde{Q}_1}{\partial z} = \frac{\partial}{\partial\hat{g}}\left[\frac{\epsilon}{\hat{g}}\right]\cdot\frac{\partial\hat{g}}{\partial z} = -\frac{\epsilon}{\hat{g}^2}\cdot\hat{g}(1-\hat{g}) = -\frac{\epsilon(1-\hat{g})}{\hat{g}}.$$

Setting $r_i^{(1)} \coloneqq y_i - \widetilde{Q}_1$ and applying the chain rule,

$$\frac{\partial}{\partial z}\Big[\beta(y_i-\widetilde{Q}_1)^2\Big] = -2\beta\,r_i^{(1)}\cdot\frac{\partial\widetilde{Q}_1}{\partial z} = 2\beta\,r_i^{(1)}\cdot\frac{\epsilon(1-\hat{g})}{\hat{g}}.$$

Differentiating once more via the product rule,

$$\frac{\partial^2}{\partial z^2}\Big[\beta(y_i-\widetilde{Q}_1)^2\Big] = 2\beta\left[\frac{\partial r_i^{(1)}}{\partial z}\cdot\frac{\epsilon(1-\hat{g})}{\hat{g}} + r_i^{(1)}\cdot\frac{\partial}{\partial z}\left(\frac{\epsilon(1-\hat{g})}{\hat{g}}\right)\right].$$

Since $\partial r_i^{(1)}/\partial z = -\partial\widetilde{Q}_1/\partial z = \epsilon(1-\hat{g})/\hat{g}$, the first term equals $\epsilon^2(1-\hat{g})^2/\hat{g}^2 \geq 0$. The second term is weighted by the residual $r_i^{(1)}$, which vanishes at convergence; dropping it yields

$$\frac{\partial^2}{\partial z^2}\Big[\beta(y_i-\widetilde{Q}_1)^2\Big] \approx 2\beta\cdot\frac{\epsilon(y_i-\widetilde{Q}_1)}{\hat{g}^2}. \tag{24}$$

Summing contributions (i)–(iii) and separating by treatment status recovers Eqs. (9) and (12):

$$\frac{\partial\ell_i}{\partial z} = \alpha(\hat{g}-t_i) - 2\beta\begin{cases} -\epsilon\,(y_i-\widetilde{Q}_0)\,\dfrac{\hat{g}}{1-\hat{g}}, & t_i = 0, \\[2ex] -\epsilon\,(y_i-\widetilde{Q}_1)\,\dfrac{1-\hat{g}}{\hat{g}}, & t_i = 1, \end{cases} \tag{25}$$

$$\frac{\partial^2\ell_i}{\partial z^2} = \alpha\,\hat{g}(1-\hat{g}) - 2\beta\begin{cases} \dfrac{\epsilon(y_i-\widetilde{Q}_0)}{(1-\hat{g})^2}, & t_i = 0, \\[2ex] \dfrac{\epsilon(y_i-\widetilde{Q}_1)}{\hat{g}^2}, & t_i = 1. \end{cases} \tag{26}$$

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